15/12/22 MD

Reg. No. | | | | | | | | | | |

[22CSE306]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER - 2022
OBJECT ORIENTED PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

PART - A (5×2=10 Marks)
ANSWER ALL QUESTIONS

1.	a.	Define the term Reusable and Extending code in OOPS.	L-1	CO1	02
	b.	State the difference between Getter and Setter Methods.	L-1	CO2	02
	c.	Summarize the use of Super Keyword.	L-2	CO4	02
	d.	Explain the need of Coupling and Cohesion.	L-2	CO5	02
	e.	Describe the term "Error" and "Exception" in Java Programming.	L-2	CO6	02

PART - B (6×5=30)

2.	a.	Distinguish Static and Final Keyword and the application of both with implementation using Java Program.	L-2	CO1	05
		OR			
	b.	Discuss the need of Object Creation in Java and the application of same with implementation using Java Program.	L-2	CO1	05
3.	a.	Demonstrate Object Oriented Architecture in Software Architecture and the Characterisitics of Object Oriented Architecture.	L-3	CO2	05
		OR			
	b.	Describe the Purpose of having Coding Standards in Software Development.	L-2	CO2	05
4.	a.	Illustrate the Types of Inheritance and discuss why Multiple Inheritance is not Supported in Java.	L-3	CO3	05
		OR			
	b.	Discuss the Multilevel Inheritance and Multiple Inheritance and the application of both with Implementation using Java Program.	L-3	CO3	05
5.	a.	Discuss the Concept of Abstraction and List the Advantages of Abstraction against Interface.	L-2	CO4	05
		OR			
	b.	Demonstrate Anonymous Inner class and recommend the need of "Anonymous Inner Class" -with realtime Scenario using Java Program.	L-3	CO4	05
6.	a.	Distinguish Static Polymorphism and Dynamic Polymorphism with realtime example and also provide the implementation of Static Polymorphism using Java Program.	L-2	CO5	05
		OR			
	b.	Demonstrate it with Implementation that which below is More Effective and Recommended? " Implementation of Dynamic Polymorphism by class inheritance or Interface ".	L-3	CO5	05

7.	a.	Distinguish the Checked Exception and Unchecked Exception	L-3	CO6	05
		OR			
	b.	Java is platfor independent. Justify. Discuss the compile and runtime of java architecture.	L-3	CO6	05

PART - C (6×10=60)

8.	a.	How Java become Platform Independent. Discuss and illustrate the compile time and runtime of Java Architecture.	L-3	CO1	10
		OR			
	b.	Illustrate the JVM Architecture and Memory Management in Java.	L-3	CO1	10
9.	a.	i. What is Package in Java and discuss the need of packages in Java Programming with an example. ii. Explain the Need of Default Constructor and Parameterized Constructor with Implementation	L-2	CO2	10
		OR			
	b.	i. What is instance variable hiding? Explain with an example. ii. Explain the need of super keyword usage in OOPS approach and Propose the need with Implementation using Java	L-2 L-2	CO2 CO2	5 5
10.	a.	Write a Java program to implement the "IS-A" Relationship and "HAS-A" Relationship	L-3	CO3	10
		OR			
	b.	Explain the need of below keywords in Inheritance and demonstrate using Java i. class ii. finally iii. new iv. Extends v. implements	L-2	CO3	10
11.	a.	Write a Java Program with below Requirement i. Create Functional Interface named "NumericTest" with abstract method "computeTest(int n)" with boolean as return type and integer as a parameter. ii. Using the above, find the even number and negative number as per parameter.	L-3	CO4	10
		OR			
	b.	Write a Java Program with below Requirement i. Create Functional Interface named "INumberGame" with abstract method "FindMissingNumber" ii. Implement the above mentioned Abstract method using Lambda Expression to Find the Missing Number of below Series. {1, 2, 4, 6, 3, 7, 8} **OR** Get the Series number from User dynamically and find the missing number using Lambda Expression	L-3	CO4	10

		Hint: Follow the naming convention of class/interface/variable/method			
12.	**a.**	Write a program in java t sort an array of ten elements. Also, find the sum of array elements using for each loop.	L-3	CO5	**10**
		OR			
	b.	Write a Java Program to implement Tight Coupling and Loose Coupling using Java Programming	L-2	CO5	**10**
13.	**a.**	Illustrate the Exception Hierarchy and demonstrate the Runtime Exception using Java program	L-3	CO6	**10**
		OR			
	b.	Write a java program to implement Custom Exception "Discount_Not_Allowed" with below condition **i.** If product_price < 500 "Discount_Not_Allowed_Exception" need to be thrown **ii.** If product_price >1000 "Discount_Not_Allowed_Exception" need to be thrown Note: Exception class Must have Constructor and "User-Defined Message"	L-3	CO6	**10**